

SMAART Minds - Technical Functionalities Document

Version: 1.0
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Platform: SMAART Minds - Student Dashboard & Learning Management System

1. Executive Summary

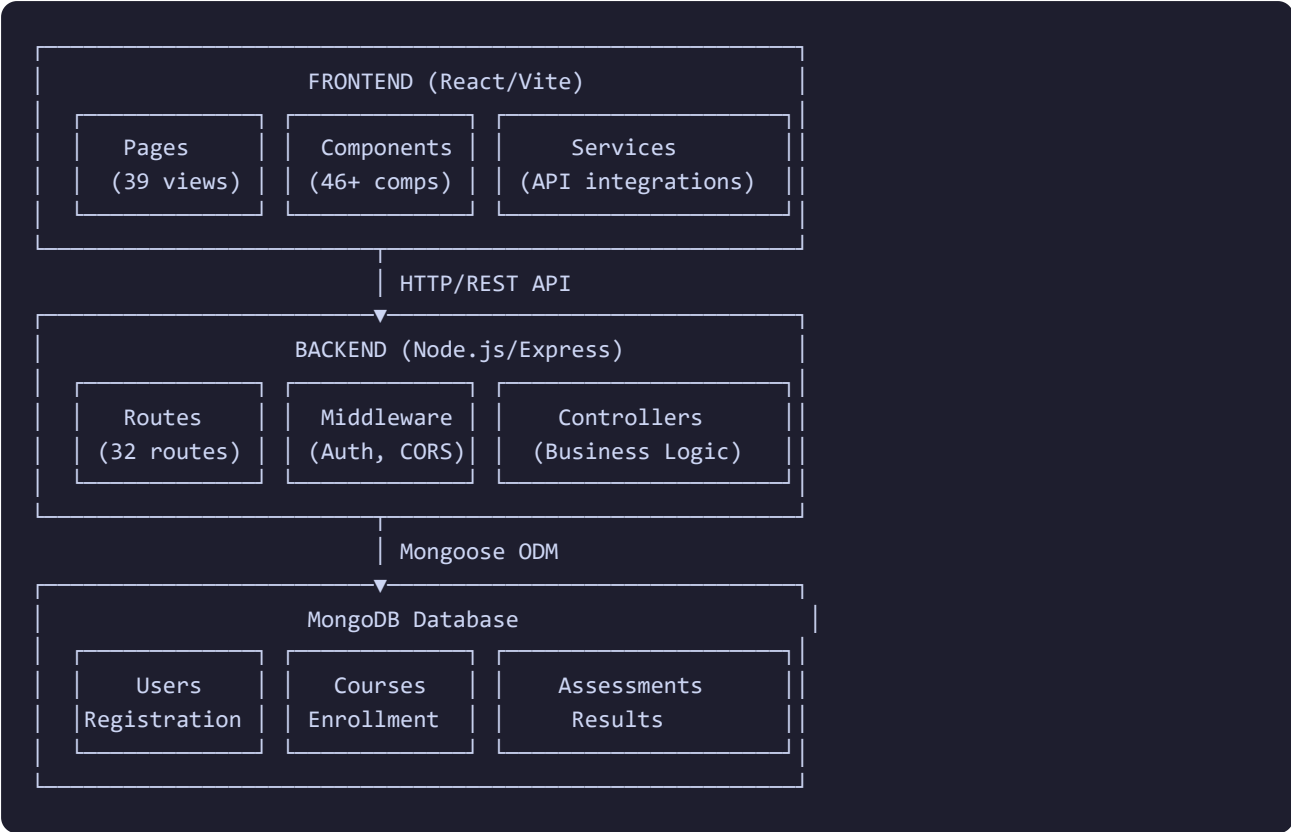
SMAART Minds is a comprehensive student learning management and development platform designed to enhance educational outcomes through personalized assessments, course management, community engagement, and holistic student development tracking.

2. System Architecture

2.1 Technology Stack

Layer	Technology
Frontend	React 18, Vite, TailwindCSS
Backend	Node.js, Express.js
Database	MongoDB with Mongoose ODM
Authentication	JWT (JSON Web Tokens)
File Storage	Local uploads + Cloudinary
State Management	React Query, Zustand
UI Components	Radix UI, Framer Motion

2.2 Architecture Diagram



3. Core Functionalities

3.1 User Authentication & Authorization

3.1.1 Features

- **OTP-Based Login**: Secure login with email OTP verification
- **First-Time Password Change**: Mandatory password change for new users
- **Role-Based Access Control**: Student, Admin, Teacher roles
- **JWT Token Management**: Secure session handling
- **Institution Selection**: Multi-tenant support for different colleges

3.1.2 Security Features

- Password hashing with bcrypt (12 salt rounds)
- JWT token expiration (24 hours)
- Rate limiting on login attempts
- Session storage for client-side tokens

3.2 User Registration System

3.2.1 Multi-Step Registration Process

Step	Section	Fields
1	Personal Details	Name, Email, Mobile, DOB, Gender, Address
2	Academic Details	Course, Department, Year, Roll Number
3	Marksheets	10th, 12th, UG/PG Semester marks
4	Certificates	Multiple certificate uploads
5	ID Proof	ID type, number, document upload
6	Password	Password creation with confirmation

3.2.2 Data Validation

- Real-time password matching indicator
- Email format validation
- Mobile number verification
- File type and size restrictions

3.3 Assessment System

3.3.1 Available Assessments

Code	Assessment Name	Questions	Purpose
ASM00001	Base Line Test	300	Initial student assessment
EQ	Emotional Quotient	Variable	Emotional intelligence
AIQ	Artificial Intelligence Quotient	Variable	AI aptitude
SQ	Social Quotient	Variable	Social skills
CQ	Creativity Quotient	Variable	Creative thinking
ARQ	Adversity Response Quotient	Variable	Resilience measurement
VAK	Visual-Auditory-Kinesthetic	Variable	Learning style

Big Five	Personality Test	Variable	OCEAN personality traits
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3.3.2 Assessment Features

- **Linear Progression:** One question at a time, no backward navigation
- **Progress Persistence:** Answers saved per-question in real-time
- **Resume Capability:** Return to exact question on revisit
- **Question Shuffling:** Randomized question order per attempt
- **Timed Tests:** Optional time limits per assessment
- **Result Analytics:** Detailed scoring and category breakdown

3.4 Course Management

3.4.1 Course Structure



3.4.2 Features

- **Video Player:** Custom video player with progress tracking
- **Course Enrollment:** Track enrolled courses per student
- **Progress Tracking:** Module-wise completion status
- **Pathway Visualization:** Interactive learning path display
- **Certificate Generation:** Automatic certificates on completion

3.5 Vision Board System

3.5.1 Features

- Create personalized vision boards
- Upload custom images and goals

- Gallery view of all vision boards
- Edit and update functionality
- Splash screen integration

3.6 Community Platform

3.6.1 Features

- **Posts:** Create, edit, delete posts
- **Comments:** Nested comment system
- **Likes/Reactions:** Post engagement metrics
- **Groups:** Community group creation and management
- **Notifications:** Badge count for new posts
- **Moderation:** Content filtering and NSFW detection

3.7 Support Ticket System

3.7.1 Ticket Categories

- Technical Issues
- Account Problems
- Course Content
- Billing Inquiries
- Feedback
- Other

3.7.2 Features

- Priority levels (Low, Medium, High)
- Status workflow (Open → In Progress → Resolved → Closed)
- File attachments (max 3 files, 5MB each)
- Admin response thread
- Rate limiting (10 tickets/hour)

3.8 Profile Management

3.8.1 User Profile Features

- Avatar customization (3D avatar system)

- Personal information display
- Academic details view
- Skills passport visualization
- Activity history tracking

3.9 MindCare Sessions

3.9.1 Features

- Coach booking system
- Session scheduling
- Mental wellness tracking
- Quiz-based assessments
- Session history

3.10 Skills Passport

3.10.1 Features

- Comprehensive skill tracking
- Visual skill representation
- Progress over time
- Export capabilities

4. Database Schema Overview

4.1 Core Collections

Collection	Purpose	Key Fields
users	User accounts	email, password, role, registrationCompleted
registrations	Registration details	personalDetails, academicDetails, documents
courses	Course catalog	title, modules, content
courseenrollments	Student enrollments	userId, courseId, progress

assessments	Assessment definitions	code, questions, scoring
results	Assessment attempts	userId, responses, score
support	Support tickets	ticketId, status, responses
communityposts	Community content	content, author, likes
visionboards	Vision boards	userId, images, goals

4.2 Result Collections

Collection	Assessment Type
baselinerequests	Base Line Test T1
eqresults	Emotional Quotient
aiqresults	AI Quotient
sqresults	Social Quotient
cqresults	Creativity Quotient
arqresults	Adversity Response
vakresults	Learning Style
big5results	Big Five Personality

5. API Endpoints Summary

5.1 Authentication Routes

```
POST /api/auth/login      - User login with email
POST /api/auth/verify-otp - OTP verification
POST /api/auth/register   - New user registration
POST /api/auth/first-login-change-password - Password reset
```

5.2 User Routes

```
GET  /api/users/:id           - Get user details
PUT  /api/users/:id           - Update user
POST /api/users/register-details - Save registration
GET  /api/users/register-details/:email - Get registration
```

5.3 Assessment Routes

```
GET  /api/assessments/:id/start - Start assessment
POST /api/results/save-answer    - Save individual answer
POST /api/results/:id/submit    - Submit assessment
GET  /api/results/user/:userId  - Get user results
```

5.4 Course Routes

```
GET  /api/courses              - List all courses
GET  /api/courses/:id          - Get course details
POST /api/courseenrollments    - Enroll in course
PUT  /api/courseenrollments/:id - Update progress
```

5.5 Ticket Routes

```
POST /api/tickets              - Create ticket
GET  /api/tickets              - Get user's tickets
GET  /api/tickets/all          - Get all (admin)
PUT  /api/tickets/:id          - Update ticket
POST /api/tickets/:id/response - Add response
```

6. Security Implementation

6.1 Authentication Security

- JWT tokens with 24-hour expiry
- HTTP-only cookie options
- CORS configuration for API access
- Rate limiting on sensitive endpoints

6.2 Data Security

- Password hashing (bcrypt, 12 rounds)

- Input sanitization
- SQL injection prevention (NoSQL)
- XSS protection
- File upload validation

6.3 Authorization

- Role-based middleware
- Resource ownership verification
- Admin-only route protection

7. Performance Optimizations

7.1 Frontend

- Code splitting with React.lazy
- Image optimization
- Caching with React Query
- Debounced search inputs

7.2 Backend

- Database indexing on frequently queried fields
- Pagination for list endpoints
- Compound indexes for complex queries
- Connection pooling

8. Third-Party Integrations

Service	Purpose
Cloudinary	Image/file storage
MongoDB Atlas	Cloud database (optional)

Lottie Files	Animations
MediaPipe	Vision/ML features
TensorFlow.js	AI features

9. Environment Configuration

9.1 Required Environment Variables

Backend (.env)

```
MONGODB_URI=mongodb://localhost:27017/minds_db
PORT=5000
JWT_SECRET=your_jwt_secret_key
NODE_ENV=development
```

Frontend (.env)

```
VITE_API_URL=http://localhost:5000
```

10. Deployment Requirements

10.1 System Requirements

- Node.js v14 or higher
- MongoDB 4.4 or higher
- 2GB RAM minimum
- 10GB storage minimum

10.2 Production Considerations

- Use MongoDB Atlas for production
- Enable HTTPS
- Set up proper CORS origins
- Implement logging and monitoring

- Regular database backups
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Document End

This document provides a comprehensive overview of the SMAART Minds platform technical functionalities. For specific implementation details, refer to the respective component documentation.